

Technical Document #669: Data Engineering - Comprehensive Overview

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Data warehouses store structured data optimized for analytical queries. They use columnar storage and support complex aggregations.

ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

Data quality ensures data is accurate, complete, and consistent. Data validation, cleansing, and monitoring are essential components.

Data lineage tracks the origin and movement of data through systems. It is essential for debugging, compliance, and impact analysis.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Key Takeaways:

- Partitioning divides data into smaller, more manageable pieces.
- Data pipelines automate the movement and transformation of data between systems.
- Stream processing handles continuous data streams in real-time.